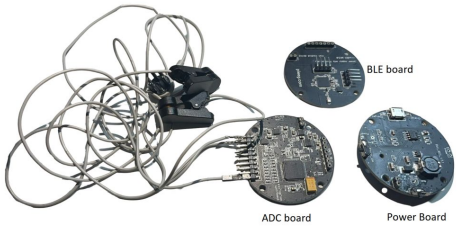
PCB & MANUFACTURING SPECIFICATION · SKU PIA24023P

Board Material & Fabrication Data

Bare-board specification, mechanical features and the production quality test for the three IronBCI-8 boards. Circuit design files are proprietary and are not part of this document.



FR-4	3 × 2	1.6 mm	Ø 50	Black
Base material	Boards × copper layers	Nominal thickness	mm round outline	Solder mask

01 BARE-BOARD FABRICATION PARAMETERS

Parameters for ordering the bare PCBs, as used by common PCB order forms (Elecrow, JLCPCB, PCBWay). Each board is ordered separately.

Parameter	Power board	ADC board	BLE board
Revision marking	V2.6	V3.2	MK3.0
Base material	FR-4, standard Tg	FR-4, standard Tg	FR-4, standard Tg
Layers	2	2	2
Outline	Ø 50.0 mm round	Ø 50.0 mm round	Ø 50.0 mm round
Thickness	1.6 mm *	1.6 mm *	1.6 mm *
Copper weight	1 oz (35 µm)	1 oz (35 µm)	1 oz (35 µm)
Solder mask / legend	Black / white	Black / white	Black / white
Surface finish	ENIG recommended	ENIG recommended	**ENIG required**
Min. track / space	0.25 mm (10 mil)	0.25 mm (10 mil)	**0.10 mm (4 mil)**
Min. drilled hole	Ø 0.49 mm	Ø 0.49 mm	**Ø 0.30 mm**
RF note	—	—	Keep-out under the chip antenna
RoHS	Lead-free	Lead-free	Lead-free

* Thickness is not stated in the fabrication data. 1.6 mm is the nominal value used here; confirm with PiEEG before ordering. The BLE board needs a fine-pitch process (QFN microcontroller).

02 LAYER STACK-UP

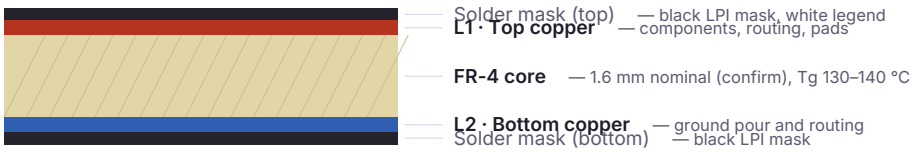


Figure 1 — 2-layer, 1.6 mm build used for all three boards. Layer roles are described in general terms only.

For research and education only. Not a medical device. Specifications subject to change.

03 MECHANICAL FEATURES

Feature	Size	Qty	Board	Purpose
Mounting holes	Ø 3.00 mm	4 each	all	M2 spacers, 29.86 × 33.04 mm pattern
Electrode header	Ø 1.00 mm plated	10	ADC	1 × 10, 2.54 mm
Board-to-board header	Ø 1.50 mm plated	7	ADC, BLE	Internal stack link
Power header	plated	2	all	GND / 5 V through the stack
SWD header	Ø 0.90 mm plated	4	BLE	Programming (ST-Link)
I²C header	plated	5	BLE	Expansion
Vias	Ø 0.49 mm	≈ 104 / 55	ADC / power	Layer transitions
Vias	Ø 0.30–0.39 mm	≈ 89	BLE	Layer transitions, microcontroller fan-out

04 ASSEMBLY

Technology	Surface-mount parts on the top side; through-hole headers and switches
Key ICs	TI ADS1299 (ADC board) · ST STM32WB55, QFN (BLE board) · LiPo charger (power board)
Connectors	1 × 10 electrode header, stacking headers, Micro USB, 2.0 mm JST, 2 slide switches
RF	2.4 GHz chip antenna at the board edge; no copper or parts in the antenna keep-out
Soldering	Lead-free reflow for SMD, selective or hand soldering for THT; no-clean flux, analog input area kept free of residue
Firmware	BLE stack and IronBCI application loaded over SWD (see IB8-SW-006)
Marking	“power supply only from battery”, “not medical device”, revision, pieeg.com

05 PRODUCTION QUALITY TEST

1 **Power and charge**
Connect a charged 200 mAh LiPo (polarity checked). With Switch 1 ON / Switch 2 OFF and USB connected, **LED 1** must light.

2 **Bluetooth**
USB unplugged, both switches ON. The device must advertise as EAREEG and connect in the Android app or PiEEG Cloud; the **BLE LED** turns on.

3 **Shorted-input check**
Short **Ref** to **channels 1–8**. Every channel must read close to 0 µV in the app; open channels show noise.

06 HANDLING, STORAGE AND COMPLIANCE

Topic	Requirement
ESD and storage	ESD-sensitive: handle by the edges on an anti-static surface. Store dry in the anti-static bag, 10–35 °C, both switches OFF, LiPo disconnected.
Environment	Indoor use only. No conformal coating; keep away from liquids and conductive gel.
Classification	Development and evaluation board for research and education, with a 2.4 GHz BLE transmitter. Not a medical device.
Design files	Schematics, layouts, Gerbers and BOMs are proprietary to PiEEG and are not published.

07 RELATED DOCUMENTS

IB8-DS-001 Product Datasheet	**IB8-MD-002** Mechanical & Assembly Drawing	**IB8-PIN-004** Connector, Electrode & Battery Guide
IB8-QSG-005 Quick Start & Safety Guide	**IB8-SW-006** Bluetooth Interface & Software Guide	